PREOPERATIVE DIAGNOSIS: Right middle lobe non-resolving pulmonary infiltrate with mediastinal lymphadenopathy in a 58-year-old female smoker.

POSTOPERATIVE DIAGNOSIS: Right middle lobe endobronchial and parenchymal inflammatory changes; transbronchial biopsy specimens obtained for histopathology and microbiological cultures.

PROCEDURE: Flexible diagnostic fiberoptic bronchoscopy, right middle lobe bronchoalveolar lavage (BAL), and fluoroscopy-guided transbronchial lung biopsy.

OPERATIVE TECHNIQUE: The patient was positioned supine in the bronchoscopy suite. Continuous monitoring of electrocardiogram, blood pressure, and pulse oximetry was maintained. Moderate sedation was achieved utilizing intravenous midazolam and fentanyl. The oropharynx was topicalized with 2% lidocaine spray. The flexible bronchoscope was introduced transorally through an Olympus bite block and passed through the vocal cords, which appeared anatomically symmetric with normal respiratory abduction.

Systematic tracheobronchial inspection revealed a structurally normal trachea, sharp carina, and patent left bronchial tree with clear segmental divisions. Inspection of the right bronchial tree demonstrated mucosal erythema and mild circumferential edema in the right middle lobe lateral and medial subsegmental orifices. No endobronchial exophytic mass or active bleeding was identified. Bronchoalveolar lavage was performed in the medial segment of the right middle lobe with 100 mL of sterile non-bacteriostatic saline instilled in 20 mL aliquots; 52 mL of cloudy effluent was recovered and submitted for cytological examination, fungal culture, acid-fast bacilli stain, and standard bacterial cultures.

Under continuous fluoroscopic guidance, flexible biopsy forceps were advanced into the lateral segment of the right middle lobe until light resistance was met, retracted 1 cm, opened, advanced, and closed during expiration. Six transbronchial biopsy specimens were procured without significant hemoptysis. Hemostasis was achieved with cold saline lavages and localized topical epinephrine (1:20,000). Post-procedure chest fluoroscopy excluded pneumothorax. The patient recovered satisfactorily in the post-anesthesia unit.